The High-Voltage Intelligent Power Module with Series-Connected Silicon Carbide MOSFETs

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Abstract-- **Silicon carbide (SiC) power devices offer superior performance over silicon couterparts, with higher switching frequencies/speeds and lower losses. The higher energy band enables SiC to achieve higher voltage rating, thus enables it to be appropriate for the high-voltage application scenarios. However, for the medium voltage(≥3.3 kV), the cost of a single chip will increase sharply compared with the main stream 1.2 kV SiC chips. To reduce the cost for the medium-voltage power module, this paper introduces a 3.3kV SiC intelligent power module (IPM) based on series 1.2 kV SiC MOSFETs. Considering that the voltage balancing between series MOSFETs is the most critical issue, the switching trajectory of the series-connected MOSFETs is comprehensively analyzed using piecewise behavior model. The steady-state voltage imbalance is minimized with passive method and the dynamic voltage imbalance is suppressed with adjustable integrated gate driver. The design process of the proposed IPM is introduced and the functionality is validated with LTspice simulation and experimental results.**

***Index Terms*-- gate driver, Power Module, Series-Connected.**

I. Introduction

SiC MOSFETs are extensively applied in various scenarios such as railway transportation, electric vehicles, photovoltaic inverter, distributed grid due to their advantages over silicon (Si) in terms of fast switching speed, high operating temperature [1]-[2]. Particularly, the higher breakdown voltage of SiC enables it to be appropriate for the high voltage applications. The maximum voltage of Si IGBT on the market is 6.5 kV while 20 kV SiC MOSFET has been reported [3].

In the high-voltage applications such as utility grid, cascaded converter topologies are usually employed to increase the voltage. Application of medium-voltage SiC power modules which are usually rated between 1.7 kV and 10 kV can greatly simplify converter topology, reduce system complexity, and improve power conversion efficiency. As claimed in [4], a 10 kV bus voltage application requires 11 cascaded levels of 1.2 kV power devices. The cascaded number can be reduced to 5 levels using 3.3 kV SiC. Using 3.3 kV SiC can reduce the cost of the gate driver, sensors, and other passives. Also, less cascaded levels can improve the system's reliability against power device failure. However, even though SiC is more appropriate for medium-voltage applications, there are not many products on the market due to the pre-mature chip fabrication technology. The main stream SiC products on the market are 650V and 1.2 kV. Thus, the price of 3.3 kV SiC MOSFETs on the market is still high. And the MV MOSFET module are still not the main-stream products on the market since the market increase is now driven by electric-mobility [5]-[7].

This paper introduces a 3.3 kV SiC intelligent power module using three series-connected 1.2 kV SiC MOSFETs. Compared with the 3.3 kV SiC dice, 1.2 kV SiC dice are the main-stream products on the market and they are cost-friendly. Therefore, it can dramatically reduce the price. However, the series-connection of SiC MOSFETs must address the voltage imbalance problem. When the device does not turn on/off synchronously, the device that turns on later and turns off earlier will withstand over-voltage which may destroy the device [8]-[11]. Therefore, it is critical to implement voltage sharing of the series-connected SiC MOSFETs.

Aiming at optimizing the voltage sharing method, the switching trajectory of the series-connected MOSFETs is derived. Based on the model, the passive method for dynamic and static voltage sharing using RC network is proposed. An adjustable signal-delay gate driver with constant supply voltage is employed. Both simulation and experimental results have validated the functionality of the proposed method.

II. Theoretical Analysis

This section describes the advantages of the SiC MOSFET and the principle of voltage imbalance in series-Connected SiC MOSFET.

A. Advantages of series SiC MOSFETs modules

In general, for a vertical MOSFET, as shown in Fig. 1. The thickness of the SiC chip is usually proportional to its on-resistance R_{dson}. This is because thinner SiC chips can reduce the path length and resistance of the current flowing through the chip, thus reducing the on-resistance. In addition, a thinner SiC chip can also improve the thermal dissipation of the device, which is helpful to reduce the junction temperature and improve the reliability. However, a higher voltage rating die needs a thicker substrate which inevitably increase the R_{dson}.

Therefore, for medium voltage SiC device, the die should be thicker and bigger which reduce the yield particularly for the SiC. Thus, the cost of a medium voltage SiC die is much more expensive than a 1.2 kV or a 1.7 kV SiC die. Based on the cost study, the cost of a 3.3 kV SiC power module with series 1.2 kV SiC MOSFET is less than 1/4 of a power module with 3.3 kV SiC dice.

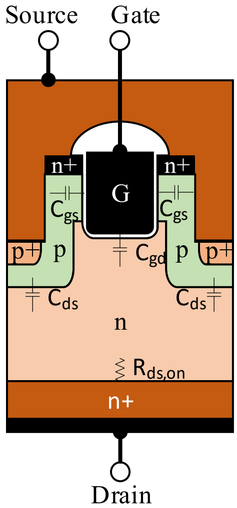


Fig. 1. The structure of a vertical MOSFET [12].

B. Mthematical model of series SiC MOSFETs voltage equalization

The analysis for multi-MOSFET connecting in series can start with two MOSFETs during the turn-off transient. The equivalent circuit of series-connected MOSFETs can be found in Fig. 2. Specifically, after the turn-on process, the MOSFETs channels are completely open and Vds reduces to normal ON voltage which is usually very low. The voltage difference on the series-connected devices is not large in ON state. Thus, this paper focuses on the turn-off process.

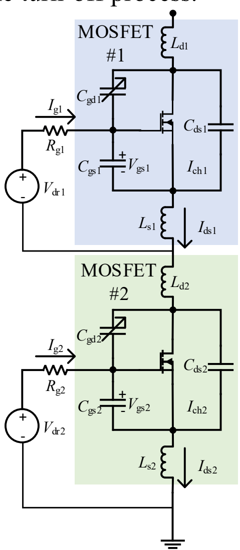


Fig. 2. The equivalent circuit of series-connected MOSFETs.

The voltage imbalance of series devices mainly includes the static unbalanced voltage when the device is off state and the dynamic unbalanced voltage during the switching transient. As shown in Fig.3. When the two SiC MOSFETs are connected in series, the static voltage formula is shown in equation (1) and (2), Where, V_{ds1_∞} and V_{ds2_∞} indicate the drain-source voltage of the upper and lower SiC MOSFETs when the SiC MOSFET reaches steady state, R_{dss1} and R_{dss2} indicate the off-state equivalent resistance of the upper and lower SiC MOSFETs, and V_{bus} indicates the bus voltage. According to (1) and (2), the static unbalanced voltage of the device in off-state is mainly caused by the mismatch of the equivalent impedance of the device in off-state, which is usually solved by the voltage equalizing resistor.

$$V_{ds1\_\infty} = \frac{R_{dss1}}{R_{dss1}+R_{dss2}} \times V_{BUS} \tag{1}$$

$$V_{ds2\_\infty} = \frac{R_{dss2}}{R_{dss1}+R_{dss2}} \times V_{BUS} \tag{2}$$

The dynamic voltage formula for the process of the device changing from on to off is shown in equation (3) and (4), V_{ds1_on} and V_{ds2_on} indicate the voltage of the SiC MOSFET when it is turned on. C_{oss1} and C_{oss2} indicate the output capacitance of the SiC MOSFET.

$$V_{ds1(t)} = V_{ds1\_\infty} + \frac{R_{dss2}V_{dss1\_on} - R_{dss1}V_{ds2\_on}}{R_{dss1}+R_{dss2}} e^{\frac{-(R_{dss1}+R_{dss2})}{R_{dss1}R_{dss2}(C_{oss1}+C_{oss2})}t} \tag{3}$$

$$V_{ds2(t)} = V_{ds2\_\infty} + \frac{R_{dss1}V_{dss2\_on} - R_{dss2}V_{ds1\_on}}{R_{dss1}+R_{dss2}} e^{\frac{-(R_{dss1}+R_{dss2})}{R_{dss1}R_{dss2}(C_{oss1}+C_{oss2})}t} \tag{4}$$

According to formulas (3) and (4), the uneven turn-off voltage during SiC MOSFET dynamics are mainly caused by the difference of device internal parameters and external parameters. The internal parameter differences include threshold voltage, output capacitance and transconductance coefficient. The external parameter differences include the driving signal delay, driving signal voltage, driving resistance and other external drive circuit parameter difference. For the differences in its own parameters, the dynamic voltage imbalance can be reduced by paralleling RC absorption circuits at the drain-source of SiC MOSFETs. For differences in external driving circuit parameters, the driving time of the single SiC MOSFET can be modified to ensure low delay of the driving signal.

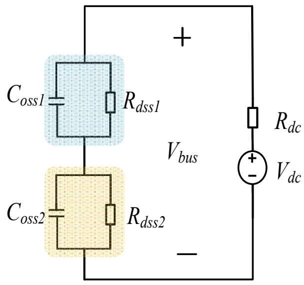


Fig. 3. Double SiC MOSFETs series turn-off equivalent model.

III. THE PROPOSED IPM

A. The circuit topology

In the module, the power circuit is shown in Fig. 4. The power circuit is divided into upper bridge arm and lower bridge arm. Each bridge arm is cascaded by three SiC MOSFETs, and six SiC MOSFETs are formed in parallel. To address the voltage imbalances caused by MOSFET intrinsic parameters, a prescreening method is applied to the MOSFETs dice to minimize parameters differences inside the module. Specifically, the gfs, R_{dson}, junction capacitance, and gate threshold voltage Vth are tested and selected. In the DBC of the module, three SiC MOSFETs in series are used to provide 3.3KV breakdown voltage, six SiC MOSFETs in parallel are used to provide 720A current. Each series-connected MOSFET is paralleled with an RC snubber circuit and a voltage-balancing resistor. To minimize the parasitic inductance between the RC network and the dice, the semiconductor-based RC chips are employed and placed on the DBC as shown in Fig. 5. The RC network can slightly reduce the dv/dt and suppress the switching oscillation in the route. Therefore, the difference on each current route is acceptable and the current sharing can be satisfied.

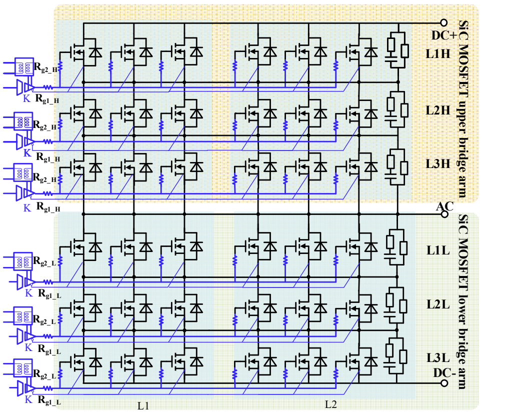


Fig. 4. The topology of the proposed series-connected MOSFET IPM.

As shown in Fig. 5. The internal DBC (directly bonded copper) structure of the module is composed of three series MOSFETs providing 3.3 kV voltage and six MOSFETs providing 720a current. When the devices are connected in series, four 15mil MOS power bonding wires are used to enhance the current stress between the upper and lower switches. A 5 Ω silicon internal gate resistor is integrated on the gate of each MOSFET. Compared with thin film resistors, silicon resistors have excellent thermal performance and are more suitable for in package applications. This resistor allows the use of hybrid (chip and wire) components with ultra-high power ratings and micro package size. Also, the layout of the direct bond copper (DBC) substrate which is shown in Fig. 5. It has been optimized to balance the parasitic inductance on each parallel route.

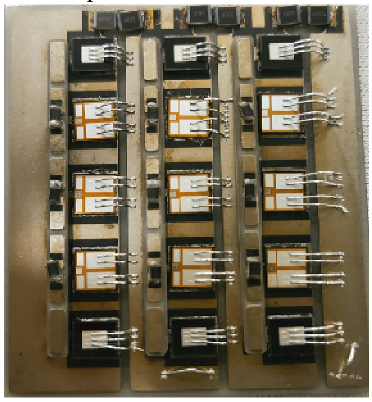


Fig. 5. DBC of the proposed series-connected MOSFET IPM.

B. The gate driver

The gate driver has three outputs, which drive the three SiC MOSFETs connected in series separately. As shown in Fig. 6. The gate driver is composed of power supply circuit, delay regulation circuit, isolation circuit and driving circuit. The power supply circuit consists of 20V DC voltage source Vdc, forward converter, rectifier circuit and voltage regulator circuit. The power supply circuit provides power for the isolation chip and the drive chip of the gate driver. The forward converter is composed of a square wave generator chip MAX13256 and a transformer. The MAX13256 chip can output the input signal to 20V square wave voltage, so that the energy can be transferred through the transformer, and then through the 1 to 1.3 boost transformer to boost it to 26V square wave voltage, and the transformer also plays the role of electrical isolation. The rectifier circuit is full bridge rectifier, using SBR05M60BLP chip to change the pulsating AC into DC. The voltage regulator circuit is composed of 18V and -5V low-voltage differential linear regulator LDO, which is used to stabilize voltage and supply power to the driver chip and isolation chip. The delay regulation circuit is shown in Fig. 6. By using the single conduction of the diode and two parallel diodes and resistors series circuit through the capacitor charging and discharging, the signal output time can be adjusted to ensure the consistency of time. The isolation circuit adopts capacitive isolation and uses NSi8210 capacitive chip. The driving circuit includes the driver chip and gate resistor Rg. The driver chip uses IXDN614, the maximum output current of the chip is 14A. The input end of the driver chip IXDN614 is connected with the output end of the isolation circuit, the output end is connected with the resistor Rg in series to output driver signal.

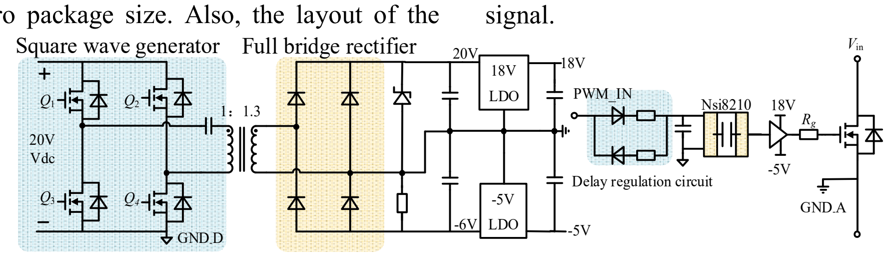


Fig. 6. The gate driver.

IV. SIMULATION AND EXPERIMENTAL VALIDATION

A. *Simulation of two SiC MOSFETs in series-connected*

In order to observe the role of RC snubber circuit in balancing dynamic voltage, LTspice is used in this paper to simulate the difference of dynamic voltage when two SiC MOSFETs are connected in series as shown in Fig. 2. The simulation results are plotted in Fig. 7, the dynamic voltage unbalance is greatly improved by parallel RC snubber circuits.

Tab. 1. The simulation setup

MOSFET	E3M0060065K	E3M0075120K
Gate signal delay	0ns	20ns
DC bus voltage	1200V	
Load current	132A	132A

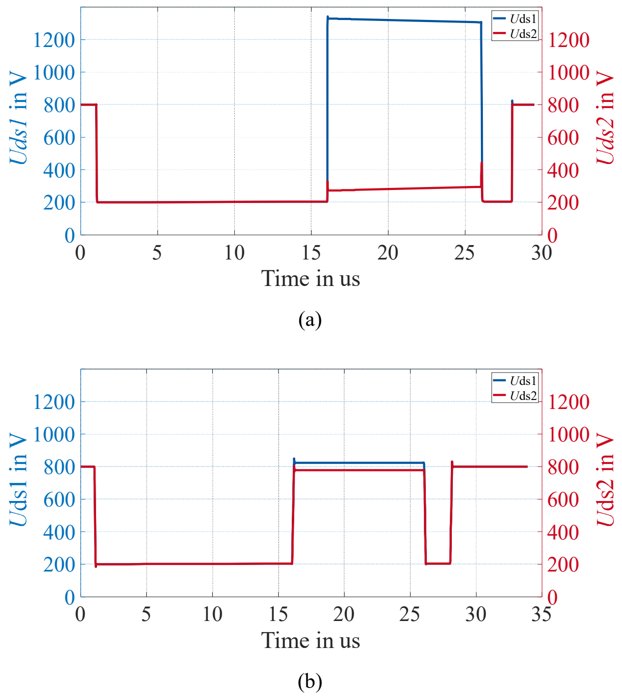


Fig. 7. Simulation results of dynamic voltage characteristics of two SiC MOSFETs in series. (a) RC absorption circuit simulation is not added. (b) RC absorption circuit simulation is added.

B. *Experiment of the gate driver*

The gate driver as described in Fig. 6. Fig. 8. is the gate driver test board. For the three series-connected MOSFETs in the bridge arm, there are three corresponding gate signal outputs. To achieve voltage balance among the three series-connected MOSFETs, it is crucial to ensure consistency in the gate driver voltage. In the experiment, considering the variations in delay among identical driver chips and the 10V/ns MOSFET chip dv/dt under nominal load current, a 5ns gate voltage delay between each series level can result in 50V voltage imbalance. Therefore, the gate voltage mismatch is controlled to be within 5ns. The output voltage waveform of gate driver is shown in the Fig. 9. The results show that the gate driver turn on and turn off delays are within 5ns. At this point, the voltage imbalance caused by gate driver delay is minimized.

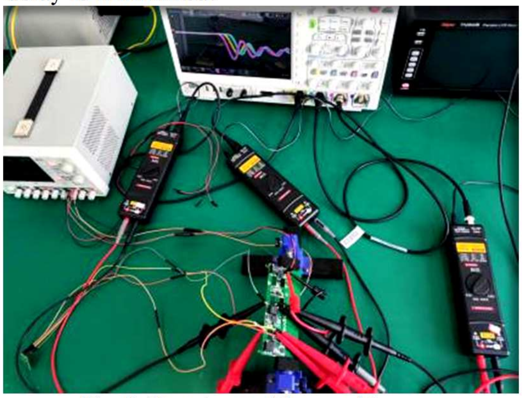


Fig. 8. Experiment of the gate driver.

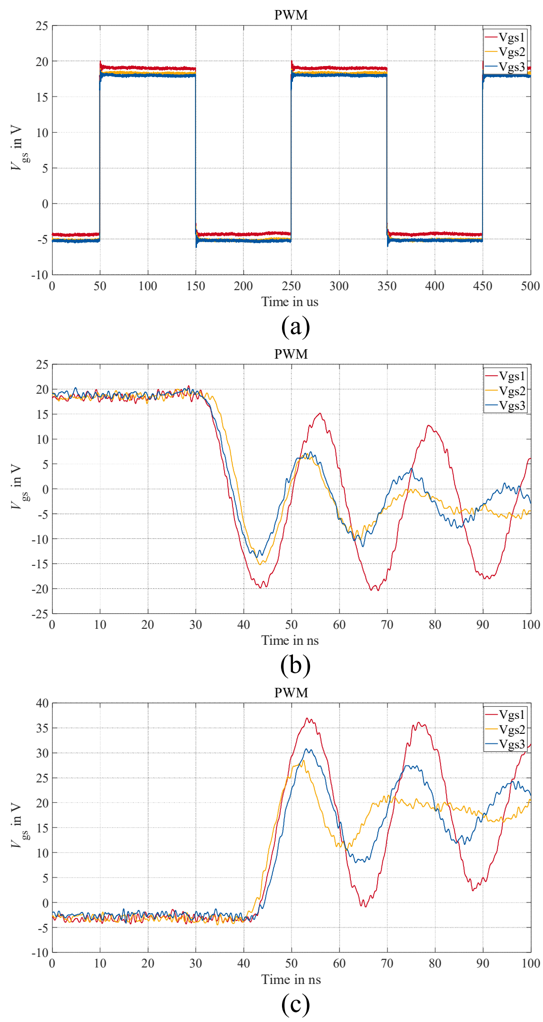


Fig. 9. The output voltage waveform of gate driver. (a) The switching frequency is 5 kHZ. (b) Amplified waveform at MOSFET turn off time. (c) Amplified waveform at MOSFET turn on time.

C. *Experiment of two SiC MOSFETs in series-connected*

In order to verify the phenomenon of voltage imbalance in series-connected MOSFETs during the actual process, as shown in Fig. 10. A double pulse test board with two MOSFETs in series-connected was designed. The double pulse test results are shown in the Fig. 11. The results show that there is voltage imbalance when different MOSFETs in series-connected, and the imbalance gradually decreases when the voltage sharing resistor in parallel-connected. Experimental results validated that the passive method based on the parallel RC-R network within the package can suppress voltage imbalance and oscillation.

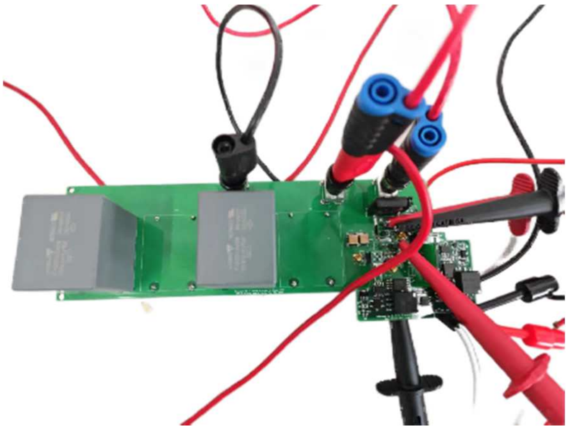
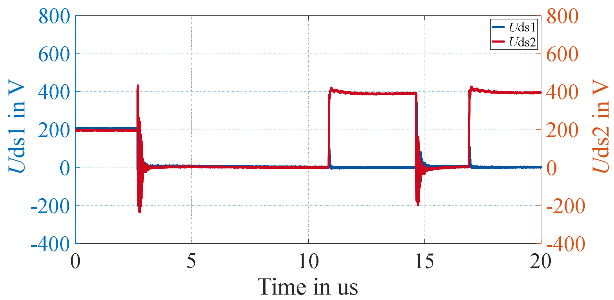
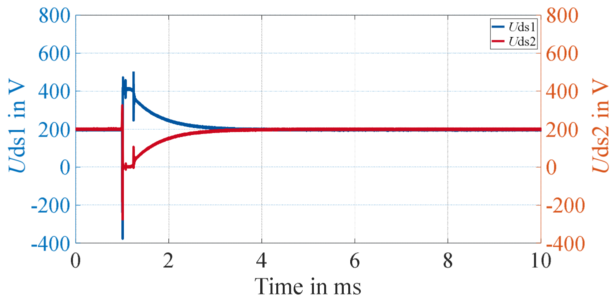


Fig. 10. Double pulse test of two MOSFET in series-connected.



(a)



(b)

Fig. 11. Results of Double pulse test of two MOSFET in series-connected. (a) Time in us. (b) Time in ms.

D. *Experiment of the IPM*

Fig. 12. shows the designed high-voltage IPM with Series-Connected Silicon Carbide MOSFETs. The power module actually achieved a rated voltage of 3.3kV. As shown in Fig. 13. The double pulse test (DPT) results have verified that the proposed module can be a cost-friendly solution for MV MOSFET module.

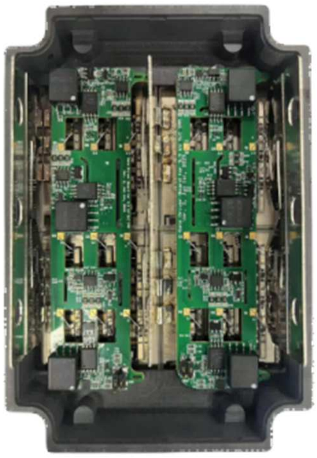


Fig. 12. Sample of the high-voltage intelligent power module with Series-Connected Silicon Carbide MOSFETs.

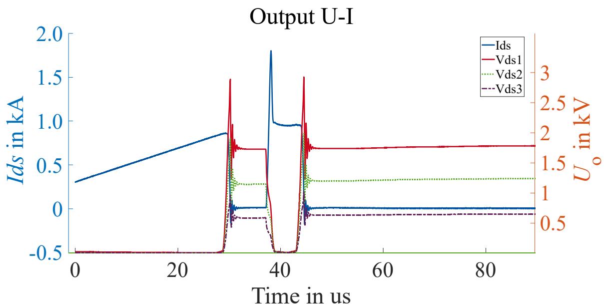


Fig. 13. Results of double pulse test of IPM.

V. Conclusion

This paper introduces an intelligent power module based on 1.2 kV MOSFETs in series, achieving a 3.3kV rating. The study delves into the application of series-connected MOSFETs in the power module, investigating the inherent mechanisms causing voltage imbalance among them. Through experimental and theoretical analysis, the paper validates the minimization of steady-state voltage imbalance through a passive approach and the suppression of dynamic voltage imbalance using an adjustable integrated gate driver. The following conclusions are drawn from the study:

(1) Voltage imbalance among series-connected MOSFETs is caused by device parameters, external circuit parameters, state parameters, and asynchronous gate signals.

(2) The use of an adjustable gate driver to adjust the drive signal delay time can suppress dynamic voltage imbalance.

(3) The integration of an RC-R networkt in parallel with the MOSFET drain-source junction suppresses voltage imbalance.

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